

Entity: mips

- File: mips.v

Diagram



Ports

Port name	Direction	Type	Description
clk	input		
reset	input		

Signals

Name	Type	Description
NPC_op	wire [1:0]	
GRF_WE	wire	
GRF_Address_from	wire [1:0]	
GRF_WD_from	wire [1:0]	
ALU_op	wire [1:0]	
ALU_B_from	wire	
DM_WE	wire	
EXT_op	wire	
currentPC	wire [31:0]	
instruction	wire [31:0]	
jal_return_address	wire [31:0]	
opcode	wire [5:0]	
rs	wire [4:0]	
rt	wire [4:0]	
rd	wire [4:0]	
shamt	wire [4:0]	
funct	wire [5:0]	
imm16	wire [15:0]	
imm26	wire [25:0]	
beq_zero	wire	
add32	wire [31:0]	

A1	wire [4:0]	
A2	wire [4:0]	
A3	wire [4:0]	
WD	wire [31:0]	
RD1	wire [31:0]	
RD2	wire [31:0]	
imm32	wire [31:0]	
A	wire [31:0]	
B	wire [31:0]	
ans	wire [31:0]	
data_input	wire [31:0]	
address	wire [31:0]	
data_output	wire [31:0]	

Instantiations

- ifu: IFU
- splitter: Splitter
- controller: controller
- grf: GRF
- ext: EXT
- alu: ALU
- dm: DM